

MODEL QUESTION PAPER

REVISION 2010

Reg. No.....

SUB CODE: 3082

Signature.....

THIRD SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING.

CHEMICAL ENGINEERING DRAWING -I.

Time:3hrs

Max.marks-100

- [Note: 1.Answer one full question from each module.
 2.A2 size drawing sheet to be used.
 3.All drawing should be in first angle projection.
 4.Both sides of the drawing sheets can be used.
 5.Dimensioning as per BIS]

MODULE-I

- 1 (a) Draw the three views of a hexagonal nut of size M25. Insert all dimension of the nut in terms of the bolt diameter.

1.Front view 2.Plan view 3. Side view (3x5)

- (b) A lap joint with a single strap for joining two plates are given with the following details

Thickness of the main plate ,t =16mm

Thickness of cover plates, t1 =16mm

Number of rows of lap rivets,nl =2

Number of rows of butt rivets,nb =1

Draw 1. The Elevation (10)

 2. The plan view (5)

OR

- II (a) Draw following three methods of locking arrangements of nuts. The diameter of the nut is 30mm.

1. Locking an ordinary nut by using a lock nut. (5)

2. Locking an ordinary nut by using a spring washer. (5)

3. Locking an ordinary nut by using a split pin. (5)

- (b) Draw sectional elevation and plan of a double riveted (zig-zag type) lap joint for joining plates

of thickness 15mm. Insert all dimensions in terms of the diameter. (10+5)

MODULE II

III. An isometric view of a socket and spigot joint is given in the fig.no.1

Draw 1. Elevation (20)

2. End view (10)

OR

IV. An isometric view of a flanged coupling(solid type) is shown in the figure No.2

Draw the bottom half sectional

1. Elevation (20)

2. End view
(10)

MODULE III

V. (a) If the detailed drawing of a unit is available, what are the steps may be adopted for the preparation of assembly drawing. (10)

(b) An isometric view of Foot step bearing is shown in the figure No.3, Draw the

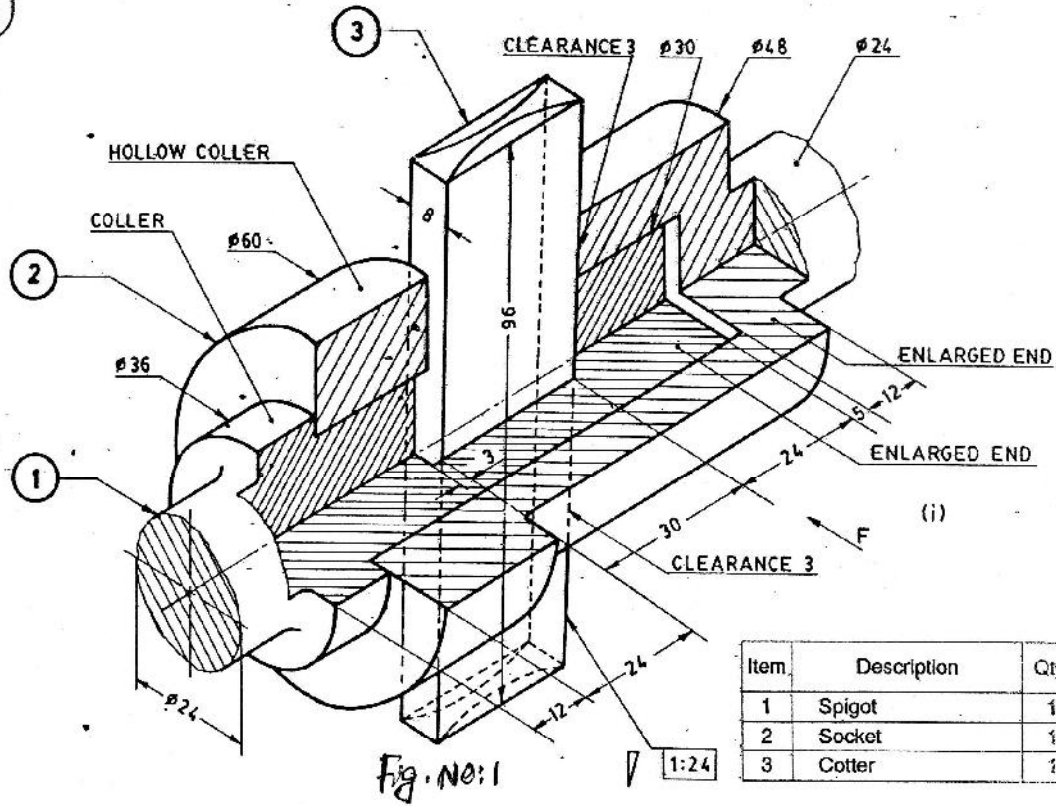
1.Elevation (20)

2.End view
(10)

OR

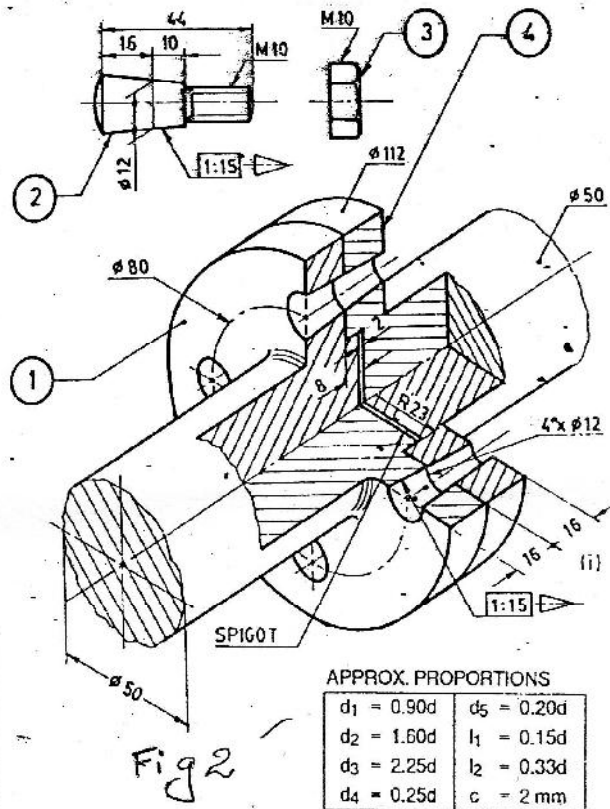
VI. Disassembled isometric view of a stuffing box is shown in the figureNo.4.Draw a top half sectional elevation and plan. (40)

①



ITEM LIST

Item	Description	Qty.	Material
1	Spigot	1	M. S.
2	Socket	1	M. S.
3	Cotter	1	M. S.



APPROX. PROPORTIONS

$d_1 = 0.90d$	$d_5 = 0.20d$
$d_2 = 1.60d$	$l_1 = 0.15d$
$d_3 = 2.25d$	$l_2 = 0.33d$
$d_4 = 0.25d$	$c = 2 \text{ mm}$

